

Exponentials & Logarithms

ANSWER KEY



Evaluating exponential expressions

- 1 Evaluate 2^5 .
 $2^5 = 32$.
 - 2 Evaluate 3^{-2} .
 $3^{-2} = \frac{1}{9}$.
 - 3 Evaluate $16^{1/2}$.
 $16^{1/2} = 4$.
 - 4 Evaluate $27^{2/3}$.
 $27^{2/3} = (27^{1/3})^2 = 3^2 = 9$.
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Evaluating logarithms

- 5 Evaluate $\log_2 32$.
Since $2^5 = 32$, $\log_2 32 = 5$.
 - 6 Evaluate $\log_3 81$.
Since $3^4 = 81$, $\log_3 81 = 4$.
 - 7 Evaluate $\log_5 1$.
 $\log_5 1 = 0$.
 - 8 Evaluate $\log_{10} 0.01$.
Since $10^{-2} = 0.01$, $\log_{10} 0.01 = -2$.
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Log laws: product and quotient

- 9 Write $\log_2 (8x)$ as a sum of logarithms.
 $\log_2 (8x) = \log_2 8 + \log_2 x = 3 + \log_2 x$.
- 10 Write $\log_3 \left(\frac{x}{9}\right)$ as a difference of logarithms.
 $\log_3 \left(\frac{x}{9}\right) = \log_3 x - \log_3 9 = \log_3 x - 2$.
- 11 Combine $\log_5 5 + \log_5 25$ into a single logarithm and evaluate.
 $\log_5 (5 \times 25) = \log_5 125 = 3$ (since $5^3 = 125$).

- 12 Combine $\log_2 24 - \log_2 3$ into a single logarithm and evaluate.
 $\log_2 \left(\frac{24}{3} \right) = \log_2 8 = 3.$
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Log laws: the power rule

- 13 Write $\log(x^5)$ using the power rule.
 $\log(x^5) = 5\log x.$
- 14 Evaluate $\log_2(4^3)$ using the power rule.
 $\log_2(4^3) = 3\log_2 4 = 3(2) = 6.$
- 15 Write $2\log_3 x + \log_3 y$ as a single logarithm.
 $2\log_3 x + \log_3 y = \log_3(x^2 y).$
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Expanding and condensing logarithmic expressions

- 16 Expand $\log\left(\frac{x^3 y}{z^2}\right)$ fully.
 $3\log x + \log y - 2\log z.$
- 17 Condense $3\ln x - 2\ln y$ into a single logarithm.
 $\ln\left(\frac{x^3}{y^2}\right).$
- 18 Expand $\log_2(\sqrt{x} \cdot y^3)$ fully.
 $\sqrt{x} = x^{1/2}$, so the expression equals $\frac{1}{2}\log_2 x + 3\log_2 y.$
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Solving exponential equations (matching bases)

- 19 Solve $2^x = 32.$
 $32 = 2^5$, so $x = 5.$
- 20 Solve $3^{x+1} = 81.$
 $81 = 3^4$, so $x + 1 = 4 \Rightarrow x = 3.$
- 21 Solve $5^{2x} = 125.$
 $125 = 5^3$, so $2x = 3 \Rightarrow x = \frac{3}{2}.$
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Solving exponential equations using logarithms

- 22 Solve $2^x = 10$ for x , rounded to three decimal places.
 $x = \frac{\log 10}{\log 2} \approx \frac{1}{0.301} \approx 3.322.$

- 23 Solve $5^x = 40$ for x , rounded to three decimal places.
 $x = \frac{\log 40}{\log 5} \approx \frac{1.602}{0.699} \approx 2.292$.
- 24 Solve $3^{2x} = 50$ for x , rounded to three decimal places.
 $2x = \frac{\log 50}{\log 3} \approx \frac{1.699}{0.477} \approx 3.561$, so $x \approx 1.781$.
- 25 Solve $e^x = 15$ for x , rounded to three decimal places.
 $x = \ln 15 \approx 2.708$.
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Solving logarithmic equations

- 26 Solve $\log_2(x) = 5$.
 $x = 2^5 = 32$.
- 27 Solve $\log_3(x - 1) = 2$.
 $x - 1 = 3^2 = 9 \Rightarrow x = 10$.
- 28 Solve $\log(x) + \log(x - 3) = 1$.
 $\log[x(x - 3)] = 1 \Rightarrow x(x - 3) = 10 \Rightarrow x^2 - 3x - 10 = 0 \Rightarrow (x - 5)(x + 2) = 0$. Reject $x = -2$ (logarithm undefined); $x = 5$.
- 29 Solve $\log_2(x + 3) + \log_2(x - 1) = 5$.
 $\log_2[(x + 3)(x - 1)] = 5 \Rightarrow (x + 3)(x - 1) = 32 \Rightarrow x^2 + 2x - 35 = 0 \Rightarrow (x + 7)(x - 5) = 0$. Reject $x = -7$ (makes $x - 1 < 0$); $x = 5$.
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Natural logarithms and e

- 30 Evaluate $\ln(e^3)$.
 $\ln(e^3) = 3$.
- 31 Evaluate $\ln(1)$.
 $\ln(1) = 0$.
- 32 Solve $\ln(x) = 2$ for x , in terms of e .
 $x = e^2$.
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Exponential growth and decay models

- 33** The population of a town is modeled by $P(t) = 5000e^{0.03t}$, where t is in years. Find $P(10)$, rounded to the nearest whole number.
 $P(10) = 5000e^{0.3} = 5000(1.34986) \approx 6749$.
- 34** A radioactive substance decays according to $A(t) = 200e^{-0.05t}$ (grams). Find $A(20)$, rounded to one decimal place.
 $A(20) = 200e^{-1} = 200(0.36788) \approx 73.6$ grams.
- 35** Determine whether $N(t) = 50e^{0.02t}$ models growth or decay, and justify your answer.
Growth — the exponent's coefficient 0.02 is positive, so N increases as t increases.
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Word problems: exponential models in context

- 36** This question has two parts. An investment of 2000 dollars grows according to $A(t) = 2000e^{0.04t}$, where t is in years.
- a** Find the value of the investment after 5 years, rounded to the nearest dollar.
 $A(5) = 2000e^{0.2} = 2000(1.2214) \approx 2443$ dollars.
- b** Find how many years it takes for the investment to double, rounded to one decimal place (using $\ln 2 \approx 0.693$). $2 = e^{0.04t} \Rightarrow 0.04t = \ln 2 \approx 0.693 \Rightarrow t \approx 17.3$ years.
- 37** This question has two parts. A cup of coffee cools according to $T(t) = 20 + 60e^{-0.1t}$ ($^{\circ}\text{C}$), where t is minutes after brewing and 20°C is room temperature.
- a** Find the temperature after 15 minutes, rounded to one decimal place (using $e^{-1.5} \approx 0.2231$).
 $T(15) = 20 + 60e^{-1.5} = 20 + 60(0.2231) \approx 33.4^{\circ}\text{C}$.
- b** Find how many minutes it takes for the temperature to drop to 35°C , rounded to one decimal place (using $\ln 4 \approx 1.386$). $35 = 20 + 60e^{-0.1t} \Rightarrow e^{-0.1t} = 0.25 \Rightarrow -0.1t = -\ln 4 \approx -1.386 \Rightarrow t \approx 13.9$ minutes.